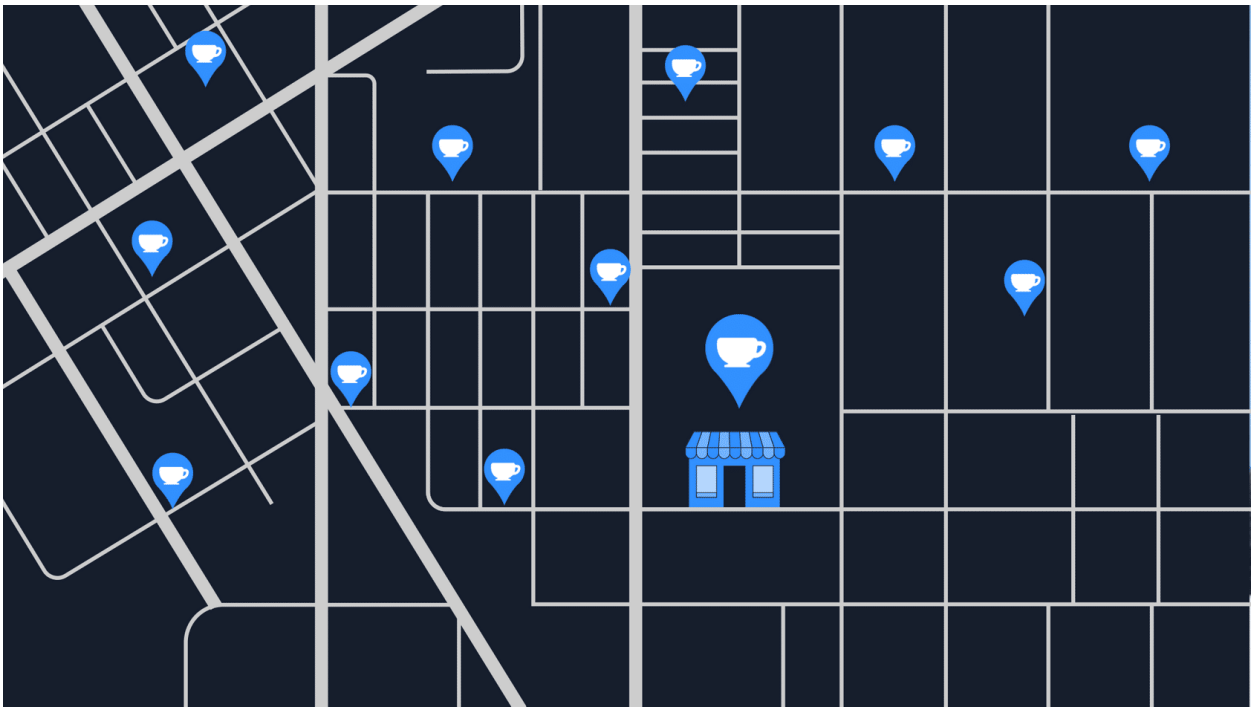
Global Infrastructure and Reliability

To understand the AWS global infrastructure, consider the coffee shop. If an event such as a parade, flood, or power outage impacts one location, customers can still get their coffee by visiting a different location only a few blocks away.



AWS Global Infrastructure

When determining the right Region for your services, data, and applications, consider the following four business factors.

- Compliance and data governance and legal requirements

  Depending on your company and location, you might need to run your data out of specific areas.

  example, if your company requires all of its data to reside within the boundaries of the UK, you would choose the London Region.

- Proximity to the customer

Selecting a Region that is close to your customers will help you to get content to them faster.

example, your company is based in Washington, DC, and many of your customers live in Singapore. You might consider running your infrastructure in the Northern Virginia Region to be close to company headquarters, and run your applications from the Singapore Region.

- Available service within a region

  Sometimes, the closest Region might not have all the features that you want to offer to customers.

  Suppose that your developers want to build an application that uses Amazon Braket (AWS quantum computing platform). As of this course, Amazon Braket is not yet available in every AWS Region around the world, so your developers would have to run it in one of the Regions that already offers it.

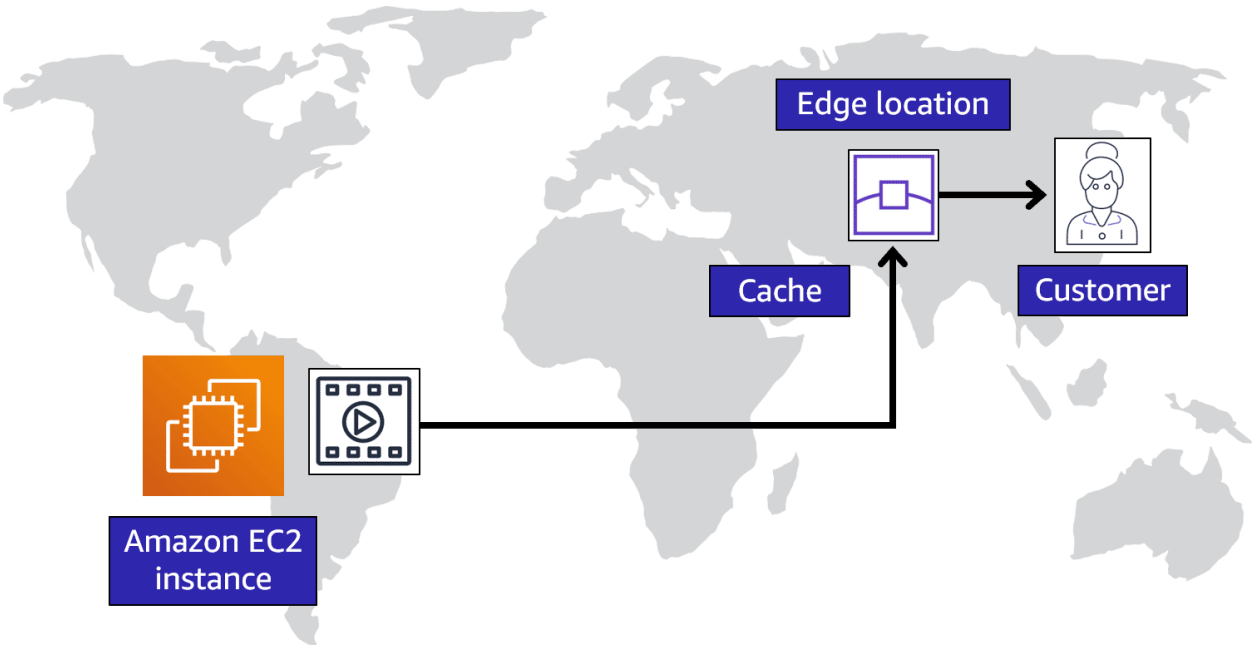
- Pricing

  Suppose that you are considering running applications in both the United States and Brazil. The way Brazil's tax structure is set up, it might cost 50% more to run the same workload out of the São Paulo Region compared to the Oregon Region.

  cost of services can vary from Region to Region.

Amazon Edge Location

site that Amazon CloudFront uses to store cached copies of your content closer to your customers for faster delivery.



> Instead of requiring your customers to get their data from Brazil, you can cache a copy locally at an edge location that is close to your customers in China.

How to Provision AWS resources

Ways of interact with AWS services

- AWS management console

  web-based interface for accessing and managing AWS services. You can quickly access recently used services and search for other services by name, keyword, or acronym.

  You can also use the AWS Console mobile application to perform tasks such as monitoring resources, viewing alarms, and accessing billing information.

  Multiple identities can stay logged into the AWS Console mobile app at the same time.

- AWS command line interface

  AWS CLI enables you to control multiple AWS services directly from the command line within one tool.

you can automate the actions that your services and applications perform through scripts.

- Software development kit

  SDKs enable you to use AWS services with your existing applications or create entirely new applications that will run on AWS.

AWS Elastic Beanstalk

provide code and configuration settings, and Elastic Beanstalk deploys the resources necessary to perform the following tasks:

- Adjust capacity
- Load balancing
- Automatic scaling
- Application health monitoring

AWS Cloud formation

treat your infrastructure as code.

build an environment by writing lines of code instead of using the AWS Management Console to individually provision resources.

enabling you to frequently build your infrastructure and applications without having to perform manual actions.

It determines the right operations to perform when managing your stack and rolls back changes automatically if it detects errors.